RIPHAH INTERNATIONAL UNIVERSITY, LAHORE CAMPUS.

SCHOOL OF COMPUTING & INNOVATION

FYP Scope and Feature Implementation

Group Member Names & SAP IDs		Project title	
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This form in intended to finalize the project features for each of groups.

FYP FEATURE DETAILS

Application Type	Feature/Sub-feature listing	Requirements
Web/ Mobile	1. User Management (Admin, Coordination, Client) 2. Role-based access control 3. Multi-criteria search (name, destination, date, budget) 4. Dashboard with personalized data 5. Emergency Help System 6. Smart Routing 7. Wishlist & Alerts 8. Virtual 3D Tour 9. Sasta Tour Packages 10. Admin Portal	1. CRUD operations for all user types 2. Role/rights assignment for each portal 3. Multi-criteria search across portals 4. Admin portal for managing app and web users 5. Offline support for emergency and routing 6. Integration with Firebase, MySQL, and APIs
AI	1. AI Chatbot for travel queries 2. Personalized Recommendations (hotels, food, routes) 3. Budget Planner with AI filters 4. Sasta Tour Package Generator 5. Smart Alerts based on weather, deals, and location 6. Virtual 3D Tour with AI-curated scenes	- Research background: NLP, personalization, federated learning - APIs used: OpenAI, Weather, Maps, Firebase - Local libraries: NLP toolkit, geolocation, caching - Code integration: Node.js backend, Flutter frontend, Firebase sync - Features support real-time decision making, offline caching, and user personalization

Sr. No	Module	User	Feature	Sub-feature	Requirement	Description
1	Authentication	Traveler, Admin, NGO	Login	Forgot Password	System shall validate credentials and allow reset via email/OTP.	Provides secure login and recovery for all users.
		Traveler, Admin, NGO	Sign Up	Email Verification	System shall send verification email during registration.	Ensures authenticity and prevents fake accounts.
2	Trip Management	Traveler	Create Trip	Destination Selection	User selects destination, dates, and preferences.	Allows travelers to plan and store trips.
		Traveler	Manage Trip	Edit/Delete Trip	User can update or cancel trips.	Provides flexibility in managing travel plans.
3	Routing	Traveler	Smart Routing	GPS Integration	System generates route with traffic/weather data.	Offers optimized travel paths with real-time updates.
		Traveler	Offline Cache	Route Storage	Routes cached locally for offline use.	Ensures continuity in low-connectivity areas.
4	Booking	Traveler	Hotel Booking	Availability Check	System fetches hotel availability via vendor API.	Provides real-time hotel options for booking.
		Traveler	Transport Booking	Bus/Train/Flight	User books transport with seat selection.	Enables seamless transport reservations.
5	Emergency Help	Traveler	SOS Alert	GPS Capture	System captures location and creates ticket.	Provides immediate emergency support with location accuracy.

		Admin/NGO	Ticket Handling	Respond to SOS	Admin/NGO view and respond to tickets.	Ensures timely response to emergencies.
6	AI Chatbot	Traveler	Query Handling	NLP Processing	Chatbot answers travel queries.	Offers conversational assistance for travel planning.
7	Recommendation Engine	Traveler	Hotel/Food Suggestion	Filters	System suggests based on budget, rating, and location.	Provides personalized recommendations for better travel experience.
8	Budget Planner	Traveler	Plan Budget Trip	Cost Calculation	System generates itinerary within budget.	Helps users plan affordable trips with AI-driven suggestions.
9	Sasta Tour Packages	Traveler	Package Generator	AI Filters	System curates cheapest possible tour plan.	Ensures affordability by generating low-cost itineraries.
10	Virtual 3D Tour	Traveler	Immersive Tour	VR/AR Support	System loads 3D assets of destinations.	Enhances user experience with immersive visuals.
11	Wishlist & Alerts	Traveler	Manage Wishlist	Save Destinations	User saves hotels/tours for later.	Allows users to bookmark preferred options.
		Traveler	Smart Alerts	Price/Weather Alerts	System sends notifications for changes.	Keeps users informed about deals and conditions.
12	Admin Portal	Admin	Manage Users	CRUD Operations	Admin adds/edits/removes users.	Provides system control and user management.
		Admin	Manage Packages	Tour Updates	Admin creates/updates tour packages.	Keeps tour content updated and relevant.

		Admin	Moderate Content	Approve/Reject	Admin reviews user-generated content.	Ensures quality and compliance of content.
13	NGO Portal	NGO	Emergency Response	Ticket Dashboard	NGO views/responds to SOS tickets.	Provides humanitarian support during emergencies.
14	Analytics & Reports	Admin	Usage Analytics	Statistics	System shows graphical stats of trips, bookings, emergencies.	Helps identify travel trends and system usage.
		Admin	Export Reports	PDF/Excel	Reports exported for meetings.	Useful for documentation and decision-making.
15	Security & Privacy	System	Data Protection	Encryption	System encrypts all user data.	Ensures privacy and security of sensitive information.
		All Users	Role-Based Access	Permissions	Access restricted by role.	Prevents misuse and enforces role-specific rights.
		Admin	Audit Logs	Activity Tracking	Logs maintained for all actions.	Provides accountability and traceability.
16	Future Enhancements	Admin	Predictive Analysis	Travel Prediction	AI predicts travel trends.	Enables preventive planning and smarter recommendations.
		Admin	ERP Integration	Vendor Alerts	Integration with external systems.	Extends system scope with external alerts.

17. Group Travel Management

User	Feature	Sub-Feature	Requirement	Description
Traveler	Join Group Trip	Accept Invitation	User can join trips through invitation links.	Enables collaborative group travel planning.
Group Organizer	Create Group Booking	Participant Management	Organizer can create group trips and invite members.	Allows management of multiple travelers in one trip.
Group Organizer	Manage Members	Add/Remove Members	Organizer controls participant list.	Ensures organized group coordination.

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18. Live GPS Tracking

User	Feature	Sub-Feature	Requirement	Description
Traveler	Enable Tracking	Location Sharing	User can share live location during trip.	Improves travel safety and visibility.
Traveler	Disable Tracking	Privacy Control	User may stop sharing anytime.	Maintains privacy and consent.
Group Organizer	Group Monitoring	Shared Live Map	System displays all active members on map.	Enables real-time group supervision.

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19. Weather Monitoring & Alerts

User	Feature	Sub-Feature	Requirement	Description
Traveler	Weather Forecast	Destination Weather	System fetches weather for trip destination.	Helps travelers plan effectively.
Traveler	Route Weather	Travel Conditions	Weather shown along selected route.	Provides journey awareness.
Traveler	Weather Alerts	Severe Weather Notifications	System generates weather warnings.	Reduces travel risks.

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20. Mental Health Support Assistant

User	Feature	Sub-Feature	Requirement	Description
Traveler	Wellness Support	Anxiety Assistance	AI chatbot provides travel-related emotional support.	Supports traveler wellbeing.
Traveler	Wellness Guidance	Recommendations	System suggests stress-management tips.	Enhances user confidence while traveling.

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21. Emergency Contact Management

User	Feature	Sub-Feature	Requirement	Description
Traveler	Manage Emergency Contacts	Add/Edit/Delete Contacts	User maintains emergency contact list.	Ensures rapid communication during emergencies.
Traveler	Emergency Notification	Contact Alert	Contacts receive location during SOS.	Improves emergency response.

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22. Notification Center

User	Feature	Sub-Feature	Requirement	Description
All Users	In-App Notifications	Booking Updates	System sends real-time notifications.	Keeps users informed.
All Users	Push Notifications	Alerts & Updates	Notifications delivered instantly.	Improves engagement and awareness.
All Users	Email Notifications	Trip & Emergency Alerts	Important updates delivered by email.	Provides multi-channel communication.

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23. Admin Emergency Dashboard

User	Feature	Sub-Feature	Requirement	Description
Admin	Emergency Monitoring	Live SOS Dashboard	System displays active emergencies.	Enables rapid response.
Admin	Emergency Map	Real-Time Location View	Admin views SOS location on map.	Supports emergency management.

User	Feature	Sub-Feature	Requirement	Description
Admin	Incident Logs	Emergency History	Stores previous emergency records.	Useful for auditing and reporting.

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24. Real-Time Synchronization

User	Feature	Sub-Feature	Requirement	Description
All Users	Live Data Sync	Firebase Sync	Changes update instantly across devices.	Maintains data consistency.
All Users	Real-Time Updates	Socket Communication	System updates without refresh.	Enhances user experience.

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25. Advanced Security

User	Feature	Sub-Feature	Requirement	Description
All Users	Multi-Factor Authentication	OTP Verification	Additional login verification layer.	Enhances account security.
System	Secure Sessions	JWT Authentication	Token-based authentication.	Protects user access.
Admin	Activity Logs	Audit Tracking	Records system activities.	Ensures accountability.